ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

MID-SEM EXAMINATION â€" DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) â€" SEMESTER 1

Time Allowed: 1.5 Hours **Maximum Marks: 30**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- PART A (DC Circuits) ---

Q.1 (a) State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL). [3 Marks]
(b) State and prove Thevenin's Theorem. Find Thevenin equivalent circuit across terminals A-B for the given resistive network. [7 Marks]
(c) State Norton's Theorem and Maximum Power Transfer Theorem. State the condition for maximum power delivery in DC circuits. [5 Marks]

[OR]

Q.1 (Alt) Apply Mesh Current analysis to determine current flowing through 10-ohm resistor in a dual-source bridge circuit. [5 Marks]

--- PART B (AC Circuits & Transformers) ---

Q.2 (a) Define RMS value, Average value, Form factor, and Peak factor for a sinusoidal alternating voltage. [5 Marks]
(b) Explain the working principle, EMF equation, and equivalent circuit of a single-phase transformer. [5 Marks]

[OR]

Q.2 (Alt) A 10 kVA, 2200/220 V, 50 Hz single-phase transformer has 88 turns on the secondary winding. Calculate primary turns, primary full load current, and maximum flux in the core. [10 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****